॥वसुधैव कुटुम्बकम्॥
SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment/ Case Study No.: 1

1. Experiment Title: Demonstrate use of inline, internal, and external JavaScript. Use console methods such as console.log(), console.error(), console.table(), and console.trace(). Create a simple webpage that logs user environment info and shows a welcome message.

2. Software/Tools Required:

- Visual Studio Code (VS Code) / Notepad++
- Google Chrome / Mozilla Firefox
- JavaScript (ES6)
- HTML5
- Windows/Linux Operating System

3. Experiment Program Code:

1. index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>JavaScript Practical 1</title>

  <!-- Internal JavaScript -->
  <script>
    function showMessage() {
      alert("Hello Aditi Rajak! This is Internal JavaScript.");
    }

    console.log("===== JavaScript Practical 1 =====");
    console.log("Student Name: Aditi Rajak");
    console.log("PRN: YOUR_PRN_NUMBER");

    console.error("This is a sample error message.");

    console.table([
      {
        Name: "Aditi Rajak",
```

```
            PRN: "YOUR_PRN_NUMBER",
            Course: "BCA"
        }
    ]);

    function traceExample() {
        console.trace("Trace Example");
    }

    window.onload = function () {

        document.getElementById("greeting").innerHTML =
        "<h2>Welcome Aditi Rajak! 👋</h2>";

        console.log("Browser:", navigator.appName);
        console.log("Platform:", navigator.platform);
        console.log("User Agent:", navigator.userAgent);

        traceExample();
    };
</script>

<!-- External JavaScript -->
<script src="script.js"></script>

</head>

<body>

<h1>JavaScript Practical 1</h1>

<p><strong>Name:</strong> Aditi Rajak</p>

<p><strong>PRN:</strong> YOUR_PRN_NUMBER</p>

<div id="greeting"></div>

<p>This page demonstrates Inline, Internal and External JavaScript.</p>

<!-- Inline JavaScript -->
<button onclick="alert('Hello Aditi Rajak! Welcome to JavaScript Practical 1.')">
Inline JavaScript
</button>

<br><br>

<button onclick="showMessage()">
Internal JavaScript
</button>
```

```
<br><br>

<button onclick="externalMessage()">
External JavaScript
</button>

</body>
</html>
```

2. script.js:

```
function externalMessage() {
  alert("Hello Aditi Rajak! This is External JavaScript.");
  console.log("===== External JavaScript =====");

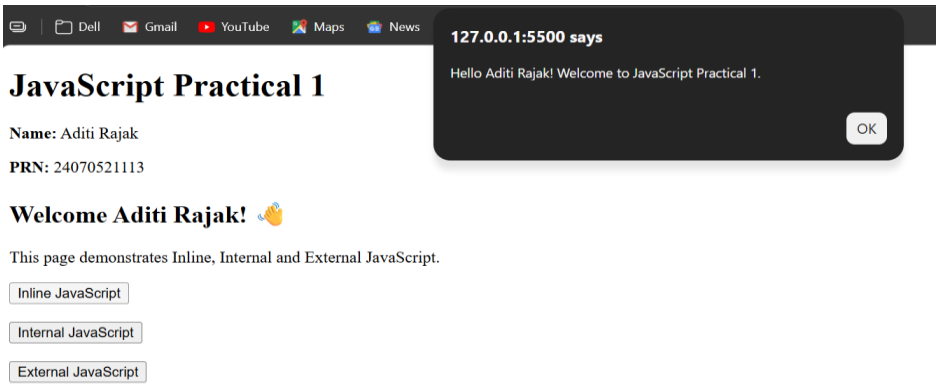
  console.log("Student Name: Aditi Rajak");
  console.log("PRN: YOUR_PRN_NUMBER");

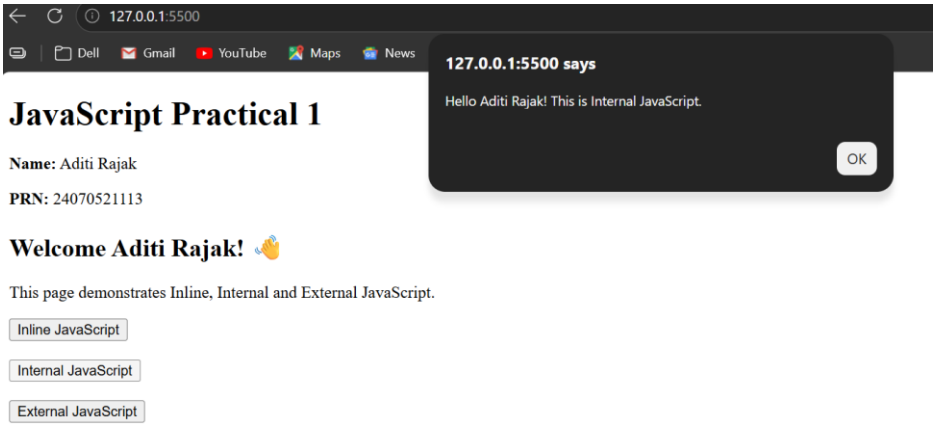
  console.log("Screen Width:", screen.width);
  console.log("Screen Height:", screen.height);

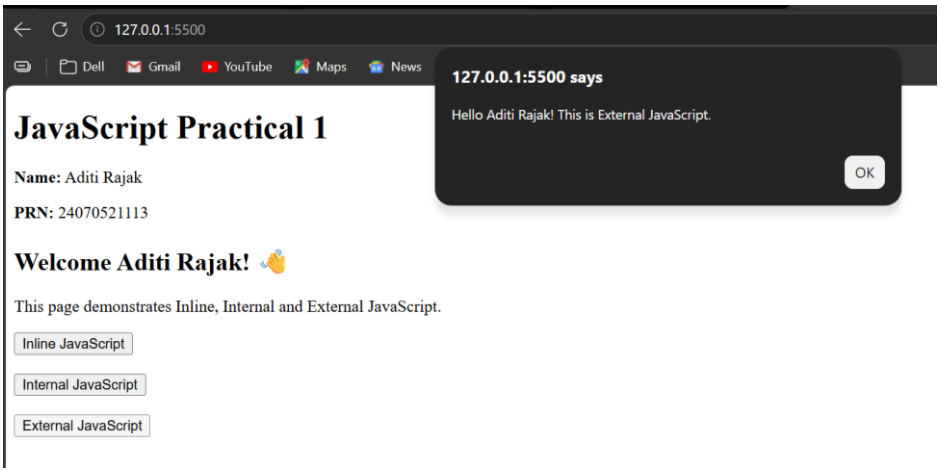
  console.log("Language:", navigator.language);
}
```

4. Output:









[JavaScript Practical 1]

5. Case Study Title: **User Registration and Welcome Portal Using JavaScript**

6. Case Study Program Code:

1. index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>User Registration Portal</title>

<style>
body{
   font-family: Arial, sans-serif;
   background:#f2f2f2;
}

.container{
   width:400px;
   margin:60px auto;
   background:white;
   padding:25px;
   border-radius:10px;
   box-shadow:0px 0px 10px gray;
}

input{
   width:95%;
   padding:10px;
   margin-top:8px;
   margin-bottom:15px;
}

button{
   padding:10px 20px;
   margin:5px;
   cursor:pointer;
}

#result{
   margin-top:20px;
   color:green;
   font-weight:bold;
}
</style>
```

```
<script>

function submitData(){

let name=document.getElementById("name").value;
let prn=document.getElementById("prn").value;

if(name=="" || prn==""){
   alert("Please enter Name and PRN");
   console.error("Input fields are empty");
   return;
}

document.getElementById("result").innerHTML=
"Welcome <b>"+name+"</b>!<br>Your PRN is <b>"+prn+"</b>.";

console.log("Student Registered Successfully");

console.table([
{
Name:name,
PRN:prn
}
]);

traceFunction();

}

function traceFunction(){
console.trace("Trace Message");
}

</script>

<script src="script.js"></script>

</head>

<body>

<div class="container">

<h2>User Registration Portal</h2>

<label>Name</label><br>
<input type="text" id="name" placeholder="Enter Name">

<label>PRN Number</label><br>
<input type="text" id="prn" placeholder="Enter PRN">
```

```
<br>

<button onclick="alert('Inline JavaScript Executed')">
Inline JS
</button>

<button onclick="submitData()">
Internal JS
</button>

<button onclick="externalInfo()">
External JS
</button>

<div id="result"></div>

</div>

</body>
</html>
```

2. script.js:

```
function externalInfo(){

alert("External JavaScript Executed");

console.log("Browser Name:",navigator.appName);

console.log("Browser Version:",navigator.appVersion);

console.log("Platform:",navigator.platform);

console.log("Language:",navigator.language);

console.log("Screen Width:",screen.width);

console.log("Screen Height:",screen.height);

}
```

- **Inline JavaScript:** JavaScript code is written directly inside an HTML element using an event attribute.
  `<button onclick="alert('Hello')">Click</button>`

- **Internal JavaScript:** JavaScript code is written inside the <script> tag in the HTML document.
  ```
  <script>
  console.log("Hello");
  </script>
  ```

- **External JavaScript :** JavaScript code is written in a separate .js file and linked using:
  <script src="script.js"></script>

- JavaScript code is written in a separate .js file and linked using:
  <script src="script.js"></script>

Console Methods Used
console.log()
Displays normal information.
console.log("Welcome");

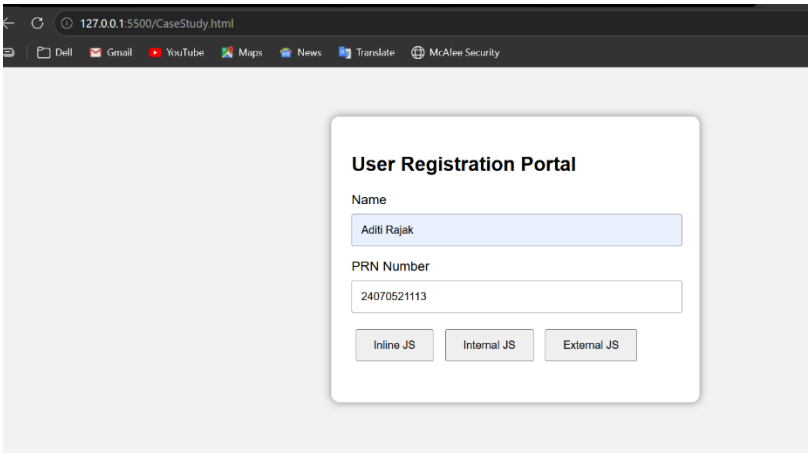
console.error()
Displays error messages.
console.error("Sample Error");

console.table()
Displays data in table format.
```
console.table([
   {Name:"Aditi",Course:"BCA"},
   {Name:"Rahul",Course:"BCA"}
]);
```

console.trace()
Shows the execution path (call stack).
console.trace("Trace Example");

7. Output:

User Registration Portal



8. Result/Conclusion: The program was successfully executed. The webpage demonstrated Inline, Internal, and External JavaScript. The console methods (console.log(), console.error(), console.table(), and console.trace()) displayed the expected output, user environment information was logged successfully, and the welcome messages were shown as intended.